

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF ECONOMICS Random
variables and discrete distributions

Week 4 · Lectures 1 and 2

Tashkent International University · Faculty of Economics

2025/2026 academic
year



From events to numbers

Weeks 1-3. We worked with events: "Drama," "R rating," "positive test." They are sets.

Sets can be added, multiplied, and conditioned. But they cannot be averaged.

A common mistake The questions we need are different.

"How many out of 10 movies are rated R?" "How many applications does a bank handle in an hour?" "How much does an insurance company pay out?"

The answer to these is a number, not an event.

Elementary event space Ω

Random variable $X: \Omega \rightarrow \mathbb{R}$

Distribution $P(X = \cdot)$

Characteristics $E[X], \text{Var}(X)$

This week the language of probability turns to arithmetic. The central limit theorem in week 7 relies on this apparatus.

Learning Objectives for Week 4

By the end of this week, you will:

1. Define a random variable as a function and distinguish it from an event.

2. Construct a probability mass function and a distribution function and explain the relationship between them.

3. Derive and calculate the expected value and variance from the definition.

4. Apply the linearity properties of E and Var .

5. Recognize the Bernoulli, binomial, and Poisson distributions and verify their conditions.

6. Evaluate the appropriateness of a model before selecting it.

Learning Outcome CLO3. Work with random variables and standard distributions, and calculate expected value and variance (Bloom: Apply).

Math requirement. This week you should work freely with the sum symbol Σ . The sum of geometric progressions is also a Challenge (MATH102).

Random amount

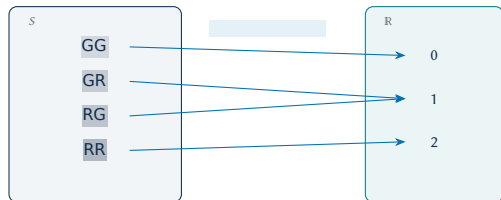
A random variable is a function

Definition A random variable is a function that maps a set of elementary events to real numbers:

$$X : S \rightarrow \mathbb{R}$$

Each outcome is assigned a single number.

A common misconception Despite its name, a random variable is neither random nor a quantity—it is a function. The randomness is in the outcome, not in the function.



We choose two movies. $\square$ — the number of R-rated movies among them. $\{\square\square, \square\square\}$ both map to $\square = 1$.

Definition Probability Mass Function

$$p(x) = P(X = x)$$

Conditions: $p(x) \geq 0$ and $\sum p(x) = 1$.

Cumulative Distribution Function

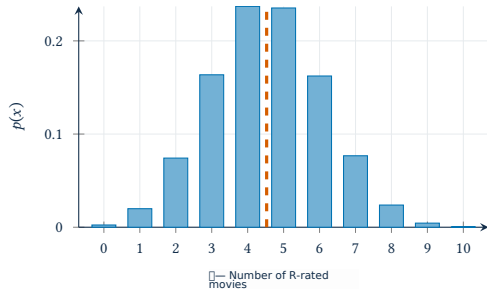
$$P(x) = P(X \leq x) = \sum_{t \leq x} p(t)$$

Properties of P :

► non-decreasing ► $P(-\infty) = 0$, $P(+\infty) = 1$ ► step-like; jump height = $p(x)$

A common error is that $P(X < x)$ and $P(X \leq x)$ are not the same in discrete cases. The difference is exactly $p(x)$. In the continuous case (week 5), the difference disappears.

Random variable from real data 2 We randomly select 10 movies from 458



X = Number of R-rated movies from the 10 selected movies.

$$P(R) = \frac{1113}{2458} = 0.4528 \text{ — this is familiar from Week 2.}$$

x	$p(x)$	$F(x)$
2	0,0742	0,0965
3	0,1636	0,2601
4	0,2370	0,4971
5	0,2353	0,7324
6	0,1623	0,8947

The most likely outcome is $X=4$, but $E[X] = 4.53$. The mode and expectation are not the same.

Mathematical expectation and variance

Definition Expected Value

$$E[X] = \sum_x x p(x)$$

This is the probability-weighted average of the values.

A common mistake is that $E[X]$ is not the “expected” value. In our example, $E[X] = 4.53$, but X is never equal to 4.53 — it is an integer.

Correct interpretation: if we repeat the experiment many times, the average of the results tends to 4.53 (law of large numbers).

Manual calculation. X : 0, 1, 2, 3; p : 0.10; 0.35; 0.40; 0.15.

$$E[X] = 0(0.10) + 1(0.35) + 2(0.40) + 3(0.15)$$

$$= 0 + 0,35 + 0,80 + 0,45 = \mathbf{1,60}$$

Linearity

$$E[aX + b] = aE[X] + b$$

$$\text{In the example: } E[3X + 2] = 3(1.60) + 2 = 6.80.$$

Dispersion: from definition to calculation formula

Definition. The mean square of the deviation from the mean:

$$\text{Var}(X) = E[(X - \mu)^2] = \sum_x (x - \mu)^2 p(x)$$

We derive:

$$\begin{aligned}\text{Var}(X) &= E[X^2 - 2\mu X + \mu^2] \\ &= E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - 2\mu^2 + \mu^2 \\ &= E[X^2] - (E[X])^2\end{aligned}$$

In the same example.

$$E[X^2] = 0 + 0.35 + 1.60 + 1.35 = 3.30$$

$$\text{Var}(X) = 3.30 - (1.60)^2 = 3.30 - 2.56 = 0.74$$

$$\text{SD}(X) = \sqrt{0.74} = 0.86$$

Even if we calculate by definition, we get 0.74. ✓

Dispersion properties

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

b vanishes: the shift does not change the dispersion. In the example:
 $\text{Var}(3X + 2) = 9(0.74) = 6.66$.

Bernoulli and binomial

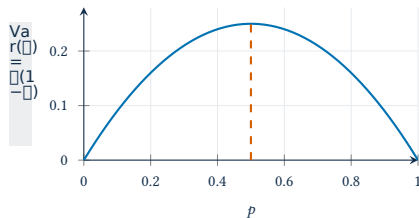
Bernoulli: the simplest random variable

Definition Bernoulli Distribution: $\square \sim \text{Bern}(\square)$ takes only the values 0 or 1.

$$p(1) = p, \quad p(0) = 1 - p$$

$$E[\square] = \square, \text{Var}(\square) = \square(1 - \square)$$

Proof. $E[\square] = 0(1 - \square) + 1(\square) = \square$.
 $E[\square^2] = 0^2(1 - \square) + 1^2(\square) = \square$, so
 $\text{Var} = \square - \square^2 = \square(1 - \square)$. $\square$



The variance is largest at $\square = 0.5$

The uncertainty is maximum at $\square = 0.5$; Since the result is almost certain at $\square \rightarrow 0$ or $\square \rightarrow 1$, the variance tends to zero.

Binomial: Bernoulli Sum

Binomial Conditions — BINS
B — two outcomes (success / failure)
I — trials are independent
N — number of trials
predetermined
S — probability of success fixed

Theorem

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = np, \text{Var}(X) = np(1-p)$$

Where did the formula come from? The probability of exactly one successful sequence in n is $p(1-p)^{n-1}$ (independent).

The number of such sequences ($\binom{n}{k}$) is the combination in week 2.

Since they are not together, the probabilities are added.

Common Mistake

$\binom{n}{k}$ — Combinatorics from Week 2. If you learned to count two weeks ago, you don't need to memorize the binomial formula.

We check the binomial on real data

$\square \sim \text{Bin}(10; 0.4528)$ theoretically:

$$E[\square] = \square\square = 10(0.4528) = 4.5281$$

$$\text{Var}(\square) = \square\square(1 - \square) = 2.4777$$

We also calculate by sum:

$$E[\square] = \sum \square\square(\square) = 4.5281 \checkmark$$

$$E[\square^2] = 22.9812$$

$$\text{Var} = 22.9812 - 4.5281^2 = 2.4777 \quad \checkmark$$

Simulation (200,000 samples, real without replacement):

$$\bar{x} = 4,5335 \quad s^2 = 2,4718$$

Very close to theoretical values.

Common mistake But we violated the condition. In sampling without replacement, the tests are not independent.

$\square - 1 = 0.9963$ — only 0.37%. If $\square < 0.05$, the binomial gives a good approximation.

Poisson distribution

Poisson: the number of rare events

Definition Poisson Distribution: $X \sim \text{Pois}(\lambda)$
— the number of events in a given interval.

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

$$E(X) = \lambda, \text{Var}(X) = \lambda$$

Poisson Conditions Events are independent; average rate does not change; no two events occur at the same time.

Common Error Note: $E(X) = \text{Var}(X) = \lambda$.

This is a testable condition. If the ratio $\text{Var}(X)/E(X)$ in the data is far from 1, Poisson is not appropriate.

Typical applications: number of customers visiting a bank branch, insurance claims, manufacturing defects, website requests.

One data set, two counts — one fits, one doesn't

NOT fits: distributors $\square$ = number of movies released by one distributor (168).

$$\bar{x} = 14,51 \quad s^2 = 1672,79 \quad \frac{s^2}{\bar{x}} = \mathbf{115,3}$$

Should be 1. Some studios release hundreds of movies, some just one — the rate is not constant.

Fits: annual Horror movies

$\square$ = number of Horror movies released in a year (2000-2018).

$$\bar{x} = 10,95 \quad s^2 = 10,39 \quad \frac{s^2}{\bar{x}} = \mathbf{0,949}$$

Very close to 1 — Poisson model is reasonable.

Audience Question Both are “counts” and from the same data set. Why does one follow Poisson and the other doesn't?

Lesson: The distribution is determined by the process condition, not the data type. Check the condition before choosing a model.

n large, p small, $np = \lambda$ constant:

$$\text{Bin}(n, p/n) \rightarrow \text{Pois}(\lambda)$$

λ P($\lambda = 3$) difference		
10	0,1172	0,0232
50	0,1386	0,0018
100	0,1396	0,0008
1 000	0,1403	0,0001
10 000	0,1404	0,00001
Pois(5) 0.1404 —		

Why is this useful? (10000 3) is hard to calculate. Poisson's formula consists of three operations.

Practical rule: If $\lambda \geq 20$ and $p \leq 0.05$, the Poisson approximation is sufficiently accurate.

Historical note Poisson (1837) derived this distribution exactly as it is — as a large number of recurrences of low-probability events.

**Economic
application**

Mathematical expectations as a decision tool

Insurance — hypothetical numbers

Conditions Annual premium: 1,200,000 soums.
Payment in case of loss: 90,000,000 soums.
Probability of loss in one year: $P = 0.008$.

Insurer's profit $\square$:

$$E[\square] = 1.2 \text{ million} - 0.008(90 \text{ million}) = 480,000 \text{ soums}$$

This is a negative expectation for the client: $-480,000$ soums.

Audience question If expectations are negative, why do people buy insurance?

Answer: dispersion. A loss of 90 million soums is catastrophic for the client. It trades a small, certain loss for a large, uncertain loss.

Expected value is not the whole story. Variance is also important—it is the basis of risk management in finance.

Risk Pooling: Why Insurance Works

Single Policy:

$$SD(\bar{X}) = \sqrt{0.008(0.992)} \times 90 \text{ million} = 8.02 \text{ million}$$

Expected 0.48 million, but variance 8.02 million — 17 times larger.

10,000 unrelated policies:

$$E = 4.8 \text{ billion, } SD = 0.80 \text{ billion}$$

$$\frac{SD}{E} = 0.167$$

Relative variance decreases by $\sqrt{n}$ times.

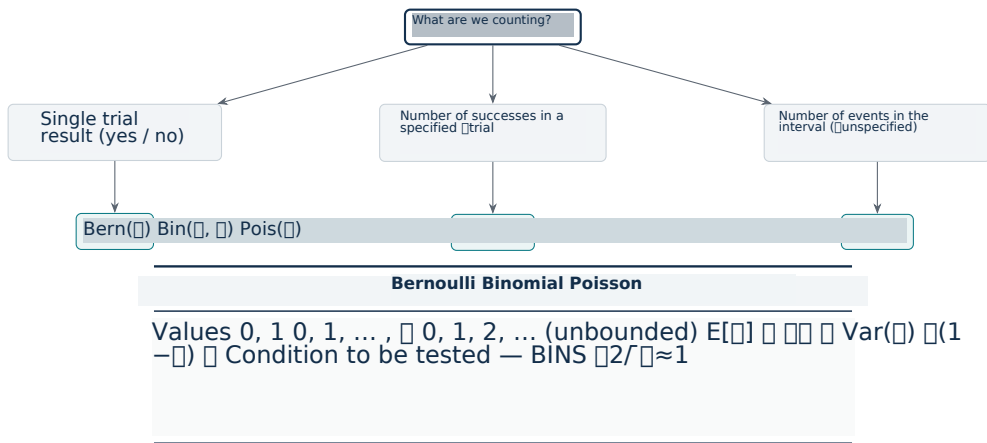
$SD / E = 16.7$ for a single policy; 0.167 for 10,000 policies — a hundred times smaller.

Common mistake Condition: policies must be independent. Pooling doesn't work if an earthquake or crisis affects all policies at the same time—one of the lessons of the 2008 financial crisis.

This $\sqrt{n}$ — is no coincidence. In week 7, it returns as $SE = \sigma/\sqrt{n}$ and becomes the basis of the entire statistical inference.

**Summar
y**

Which distribution should we choose?



Common student errors and misconceptions

Misconception Why it is wrong

" $E[\cdot]$ is the most likely value." In our example, the mode is $\cdot = 4$, the expectation is 4.53. $E[\cdot]$ is the weighted mean, not the mode; it could also be a value that is not possible at all.

"The count is the data, so it is Poisson." In the distribution of counts, $\cdot^2/\cdot = 115$. The distribution is determined by the process condition, not the type of data.

" $\text{Var}(\cdot + \cdot) = \text{Var}(\cdot) + \cdot$." Shifting does not change the variance: $\text{Var}(\cdot\cdot + \cdot) = \cdot^2\text{Var}(\cdot)$. $\cdot$ simply disappears.

"The binomial can be applied to any $\cdot$ tests." BINS conditions are required. In non-regression sampling, independence is violated — approximation is acceptable only if $\cdot < 0.05\cdot$.

" $P(\cdot < 3)$ and $P(\cdot \leq 3)$ are the same." In the discrete case, the difference is $\cdot(3) = 0.1636$ — not a small number.

"If the expectation is positive, the decision is correct." In insurance, the expectation is negative for the customer, but the decision is reasonable — because the variance is also important.

"It is not necessary to check $\sum \cdot(\cdot)$." This is the single fastest way to find the error. If it is not equal to 1, there is an error somewhere.

Main conclusions

1. A random variable is a function. It converts events into numbers and paves the way for arithmetic.

2. $E[X] = \sum x P(X=x)$ and $\text{Var}(X) = E[X^2] - (E[X])^2$ — both must be derived, not memorized.

3. E is linear, but Var is not: $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

4. Binomial — (n, p) (week 2) and independence (week 2) together.

5. Before choosing a distribution, check the condition: BINS or $np \approx 1$.

6. Expectation is half of the decision; the other half is variance. The $\sqrt{n}$ law returns in week 7.

Next week is week 5 — Continuous distributions and normal distributions. We move from sum to integral: density function, uniform and exponential distributions, normal distribution, $\hat{\mu}$ -estimate, and Q-Q plot.

This week's assignments

Assignment Content Due Date

Required Reading Newbold Chapter 4; McClave Chapter 4; Heumann Chapter 7 Before Lecture 1 Lab W04-lab — Probability Mass Function, Binomial, and Poisson Conformance Check In Lab

Homework W04-homework — 6 problems, 40 points Week 5, Monday Problem Set 3 MT3 (discrete distributions) Due: Week 5 Project Stage 4 Obtaining information — Control point 1. Source, volume, method of obtaining

Week 5, Friday

SI Assignment Find the errors in SI's choice of distribution Week 5, Friday

Project control point. This stage is not graded, but is mandatory: if you do not have the information, the project will stop from week 6. If you cannot find the source, contact the teacher immediately.

1. Newbold, P., Carlson, W. & Thorne, B. Statistics for Business and Economics. Pearson. Chapter 4 — Discrete Random Quantities and Probability Distributions.
2. McClave, J. T. & Sincich, T. (2018). Statistics, 13th ed. Pearson. Chapter 4.
3. Heumann, C., Schomaker, M. & Shalabh (2016). Introduction to Statistics and Data Analysis. Springer. Chapter 7 — Random Quantities; Chapter 8 — Distributions.
4. DeGroot, M. & Schervish, M. (2012). Probability and Statistics, 4th ed. Chapters 3–4 (expanded).
5. Poisson, S. D. (1837). Recherches sur la probabilité des jugements — the original source of distributions.
6. Wooldridge, J. M. (2019). Introductory Econometrics, 7th ed. Appendix B — Probability for Economists.
7. Interactive: Seeing Theory, Brown University, “Probability Distributions” section — seeing-theory.brown.edu
8. Video: MIT OCW 18.05, lectures 5–7; StatQuest, “Binomial Distribution” and “Poisson Distribution”.